

Darwin Meets Deep Learning:

Evolving Neural Agents for Survival in Adversarial Environments

A Neuro-Evolutionary Approach to Emergent Adaptive Behavior

Krishna Barai

Department of Electrical Engineering
Indian Institute of Engineering Science and Technology, Shibpur
Durgapur, India
2024eeb052.krishna@students.iiests.ac.in

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Abstract

This study demonstrates how neuro-evolution—combining evolutionary algorithms with neural network controllers—enables autonomous agents to develop survival strategies in a complex, adversarial 2D environment. Prey agents governed by simple feedforward neural networks evolved over generations to optimize navigation, foraging, predator avoidance, and kin-coordination behaviors. Using a genetic encoding of network parameters (weights, biases, output scaling) and fitness-driven selection, agents consistently developed emergent spatial navigation patterns, group cohesion, and risk-reward balancing without gradient-based training. Results reveal selective sweeps reducing genetic diversity, spatial heatmaps illustrating evolved migration corridors, and asymmetric co-evolutionary dynamics between prey and predators. The system provides a scalable framework for evolving adaptive behaviors in decentralized swarm systems.

1 Introduction

The integration of Darwinian evolutionary principles with neural control architectures represents a paradigm shift in artificial intelligence. While gradient-based optimization dominates deep learning, neuro-evolution offers a biologically inspired alternative for training adaptive agents in environments where reward signals are sparse or multi-objective. This work bridges evolutionary biology and machine learning by evolving populations of neural agents in a dynamic 2D world featuring:

- Spatial resource gradients (food-rich vs. hazardous zones)
- Mobile predators
- Physical obstacles
- Kin-group dynamics

The core hypothesis posits that selection pressure alone can produce complex, spatially organized behaviors in decentralized agents, mirroring natural phenomena like migration, flocking, and risk-

averse foraging. Unlike reinforcement learning, agents here have no external reward signal; survival itself drives adaptation through a multi-component fitness function.

2 Methods

2.1 Codebase Overview

The entire simulation is implemented as a single Jupyter notebook (`evo2d.ipynb`), publicly available at <https://github.com/HeroicKrishna160905/Neuro-Evolution>. The code leverages `numpy` for computation and `matplotlib` for visualization, and is structured for clarity and extensibility.

2.2 Environment Specification

A square arena (-10.0 to 10.0 on both axes) features:

- **Food patches:** Randomly distributed on the right side ($x > 0$)
- **Obstacles:** Impassable vertical walls at $x \approx -4.5$ and $x \approx +4.5$
- **Hazard zones:** Poisonous areas in the center (penalty: -1.0 fitness)
- **Predators:** Agents pursuing nearest prey at each timestep (removing prey within critical distance)

Prey agents lose energy per timestep and replenish it by consuming proximate food.

2.3 Agent and Neural Controller Architecture

Each agent is controlled by a minimalist feedforward neural network. Genomes are flat NumPy arrays encoding all weights, biases, and output scalings. Network decoding (from code):

```
# Genome decoding (as defined in source code)
idx = 0
self.w = genome[idx:idx+self.input_size*self.output_size].reshape((self.input_size, self.output_size))
idx += self.input_size*self.output_size
self.b = genome[idx:idx+self.output_size]
idx += self.output_size
self.out_scale = np.abs(genome[idx:idx+self.output_size]) + 0.2
out = np.tanh(np.dot(x, self.w) + self.b) * self.out_scale
```

Inputs:

1. Normalized 2D position
2. Direction vector to nearest food
3. Count of nearby kin (radius-based)
4. Average signal from nearby kin

Outputs:

1. Movement vector (dx, dy)
2. Broadcast signal $[-1, 1]$

2.4 Fitness Function

Agent fitness combines multiple objectives:

$$\mathcal{F} = \underbrace{1.0}_{\alpha} \times (\text{food}) + \underbrace{0.25}_{\beta} \times (\text{survival}) + \underbrace{0.1}_{\gamma} \times (\text{energy}) + \underbrace{0.5}_{\delta} \times (\text{kin proximity}) - \underbrace{1.0}_{\varepsilon} \times (\text{predator penalty})$$

This balances energy acquisition, longevity, risk aversion, and kin cooperation.

2.5 Evolutionary Algorithm

- **Selection:** Roulette-wheel (fitness-proportional) with 5% random “drift”
- **Reproduction:**
 - Uniform crossover (gene-wise parent selection)
 - Gaussian mutation (per-gene perturbation)
- **Lineage Tracking:** Child color is an average of parents’ colors for visual ancestry.
- **Generational Loop:**
 1. Population evaluation (fixed timesteps)
 2. Fitness computation
 3. Parent selection
 4. Crossover + mutation $\rightarrow$ new population

2.6 Visualization and Metrics

Metrics and figures are generated via `matplotlib` and exported as PNG/GIF (e.g., `pop_gen999.gif`).

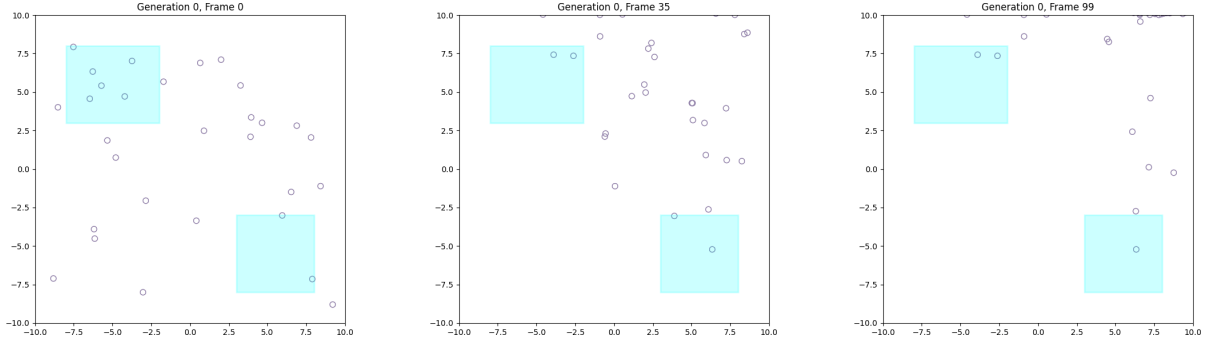
The code tracks and plots:

- Best vs. average fitness per generation
- Genome diversity (weight variance)
- Behavioral spread and heatmaps of agent density
- GIF animations of swarm evolution

3 Results

3.1 Fitness Progression

Agents exhibited rapid improvement in survival strategies over 1,000 generations.



(a) Frame 1

(b) Frame 25

(c) Frame 100

Figure 1: Snapshots from the simulation of generation 1000, showing the population at frames 1, 25, and 100. See code and figures at [GitHub repository](#).

3.2 Genetic Diversity Dynamics

Selective sweeps reduced genome diversity as optimal controllers dominated.

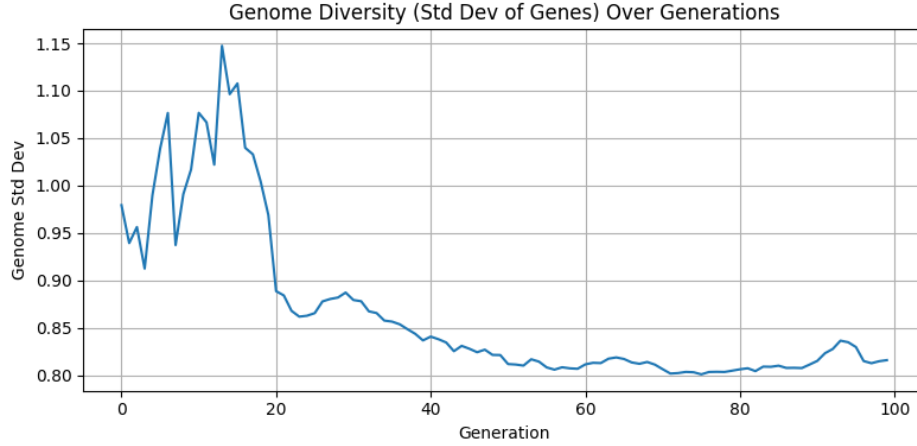


Figure 2: Average pairwise genome distance over generations.

3.3 Emergent Spatial Navigation

Position heatmaps reveal the evolution of instinctual migration corridors.

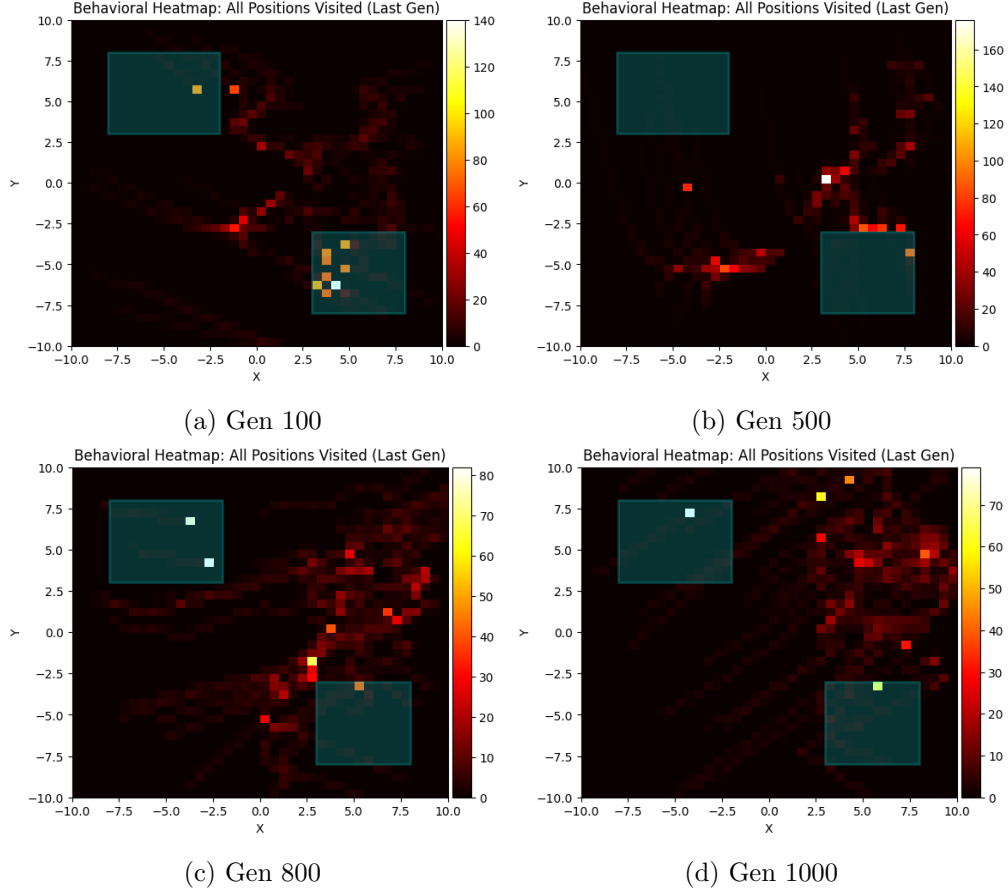


Figure 3: Spatial density heatmaps reveal migration patterns and hazard avoidance.

3.4 Kin-Coordination and Signaling

Agents evolved implicit flocking behavior:

- Kin proximity bonus ($\delta = 0.5$) promoted clustering
- Broadcast signals facilitated crude information sharing
- Genetically similar agents exhibited identical movement strategies

Collective migration emerged without explicit coordination, resembling biological canalization.

3.5 Co-Evolutionary Arms Race

When predators (identical neural architecture) evolved concurrently:

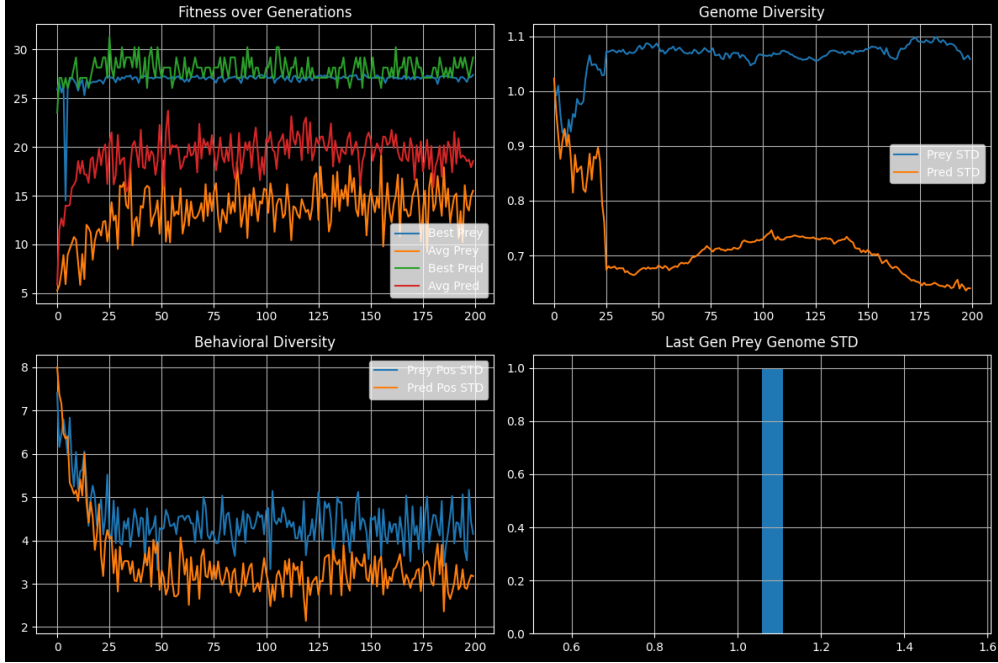


Figure 4: Co-evolutionary dynamics: prey vs. predator fitness, diversity, and tactics.

Key Observations:

- Prey fitness consistently outpaced predators
- Predator diversity collapsed early; prey maintained higher variability
- Emergent tactics included herding, flanking, and risk-averse foraging

4 Discussion

4.1 Emergent Biological Analogues

The system spontaneously replicated natural phenomena:

- **Migration Corridors:** Optimized paths resembled wildebeest/bird migration
- **Canalization:** Genetic convergence produced consistent instinctual behaviors
- **Kin Selection:** Kin proximity bonuses promoted family-group survival
- **Selective Sweeps:** Dominant genotypes rapidly outcompeted others

4.2 Neuro-Evolutionary Mechanics

Key Success Factors:

- Fitness function balance (α/ε tradeoff prevented over-specialization)
- Spatial constraints accelerated obstacle navigation evolution
- Minimalist networks (no hidden layers) sufficient for complex behaviors
- Broadcast signals enabled proto-communication despite 1D limitation

Limitations:

- Shallow networks restricted strategy complexity

- Static environment reduced open-ended adaptation
- No predator evolution in initial experiments

5 Conclusion and Future Directions

This work demonstrates neuro-evolution as a potent framework for autonomously generating adaptive behaviors in adversarial environments. Key achievements include:

1. Evolution of collision-free navigation in cluttered spaces
2. Emergence of flocking and signaling without explicit coordination
3. Co-evolutionary arms races between prey/predators

Future Work:

Extension	Potential Impact
Richer sensory inputs	Enhanced terrain awareness
Predator co-evolution	More dynamic arms races
Recurrent neural controllers	Memory-based strategies
Dynamic resource allocation	Realistic ecological modeling
Robotic swarm implementation	Physical validation of emergent behaviors

The accompanying codebase enables further exploration of these directions. Neuro-evolution proves particularly promising for scenarios where reward functions are multi-objective, environments are partially observable, and centralized control is infeasible.

Supplementary Material and Data Availability

All code, simulation outputs, figures, and animations related to this study are available in the public GitHub repository: <https://github.com/HeroicKrishna160905/Neuro-Evolution>

This includes:

- `evo2d.ipynb`: Complete simulation notebook
- `pop_gen999.gif`: Evolution animation across generations
- `heatmap_800.png`, `heatmap_1000.png`: Spatial density heatmaps

For an extended walkthrough and narrative explanation of the project—including additional figures and interpretation—readers may refer to the Medium article: <https://tinyurl.com/neuralevo>

All resources are open-source and freely accessible for further exploration and reproduction.

References

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